

5.3 Information Sources (Primary, Secondary, Tertiary)

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 08 of 12

★ Golden

5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

LEARN IT

Information sources are ways to find knowledge and data. The internet has vast information, but **not all of it is accurate**. Sources are divided into three main categories.

1. Primary sources give **original, firsthand information** straight from people, events or official records. They are **highly authentic** (genuine).

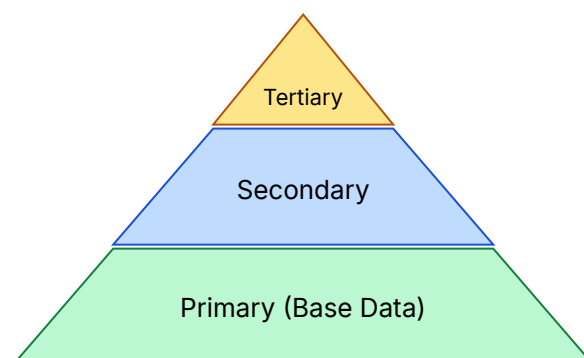
Examples: interviews, photographs, diaries, NADRA official records.

2. Secondary sources contain information that has **already been collected and explained by someone else**. They **summarise or analyse primary data**.

Examples: textbooks, news reports, research papers, Pakistan Bureau of Statistics reports.

3. Tertiary sources provide **highly summarised and organised information** derived from both primary and secondary sources. They are used for **quick reference**.

Examples: encyclopedias, Wikipedia, dictionaries, handbooks.



The information pyramid: primary data is the base; each higher level is built from the levels below it and is more summarised.

Many examples, sorted

Primary (firsthand)	Secondary (explained by others)	Tertiary (summarised, quick reference)
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Interview with a flood victim in Dadu	Newspaper report on the 2022 floods	Encyclopedia entry "Floods in Pakistan"
Photograph taken at an event	Textbook chapter explaining the event	Wikipedia article
Personal diary or letter	Biography written by a historian	Dictionary
NADRA official records (CNIC data)	Pakistan Bureau of Statistics report analysing census data	Handbook / almanac of facts
Original survey responses	Research paper analysing the survey	Index or list of research papers
Sensor readings from an IoT weather station	TV news analysis of the weather	A school "fact sheet" on climate
Quaid-e-Azam's original speech recording	Article discussing the speech	Encyclopedia summary of his life

When do sources become unreliable?

In the digital age you must check facts carefully. A source is **unreliable** (cannot be trusted) if:

Source type	Reasons for unreliability
Primary	Information is incomplete, data is recorded incorrectly, or false information was deliberately provided.
Secondary	The information is outdated, lacks proper references/citations, or has not been verified by an expert.
Tertiary	Comes from an unknown/unverified website, or the information is oversimplified and misses key facts.

REMEMBER IT

Rule of thumb (Author – Date – References): If an online article does not state the **author's name**, the **date of publication**, or list its **references**, it should not be trusted for academic work. Remember "**ADR**" — like checking a bus has a driver, a route and a ticket before you board.

WORKED EXAMPLE

Classify these — worked solutions:

(a) Your grandfather's handwritten diary from 1947. Written by a person who lived through the event → firsthand → **Primary**.

(b) A Dawn newspaper article analysing the budget. A journalist collected facts and explained them → **Secondary**.

(c) An Oxford English–Urdu dictionary. Organised, summarised reference built from many sources → **Tertiary**.

(d) A Bureau of Statistics report on literacy in Sindh. It analyses census (primary) data → **Secondary**.

(e) A photo you took of a road accident. Original record of the event → **Primary**.

Method: Ask (1) "Was the creator *there* / is it the original record?" → Primary. (2) "Is someone *explaining or analysing* original data?" → Secondary. (3) "Is it a *quick-reference summary* of many sources?" → Tertiary.

CHECK YOURSELF

Exercise — classify these as Primary (P), Secondary (S) or Tertiary (T):

1. A video interview with a cricketer after a match.
2. Your Computer Science textbook.
3. Wikipedia page on "Mohenjo-daro".
4. A birth certificate issued by the union council.
5. A research paper reviewing ten studies on smog in Lahore.
6. A handbook of physics formulas.
7. Original exam answer sheets of students.
8. A TV news report summarising a court decision.

Answers: 1. P 2. S 3. T 4. P 5. S 6. T 7. P 8. S

COMMON MISTAKE

Mistake: Thinking "primary" means "most important" or "tertiary" means "useless". **Fix:** The names describe *how far the information is from the original event*. Primary = straight from the source; tertiary = summarised summary. Each is useful for a different job. Also, Wikipedia is **tertiary** — not primary — because it summarises other sources.

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔ پرائمری ذرائع براہ راست اور اصل معلومات دیتے ہیں (جیسے انٹرویو، تصاویر، نادرا کا ریکارڈ)۔ سیکنڈری ذرائع وہ ہیں جن میں کسی اور نے پرائمری معلومات کی وضاحت یا تجزیہ کیا ہو (جیسے درسی کتابیں، خبریں)۔ ٹرشییری ذرائع خلاصہ شدہ معلومات دیتے ہیں جیسے انسائیکلوپیڈیا، وکی پیڈیا اور لغت۔ جس آن لائن مضمون میں مصنف کا نام، تاریخ یا حوالہ جات نہ ہوں اس پر بھروسہ نہ کریں۔

EXAM TIP

Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short), and "When does a source become unreliable?"

Write the definition key phrase for each — *firsthand/original, collected and explained by someone else, highly summarised for quick reference* — plus two examples each. Draw the pyramid with primary at the base. For unreliability, give one reason for each type and quote the author–date–references rule.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A primary source is:

- a. Original, first-hand
- b. A summary of summaries
- c. Always Wikipedia
- d. A pie chart of rumours

MCQ 2. A secondary source is:

- a. Someone else's analysis of primary material
- b. Raw sensor logs only
- c. Always false
- d. A NAND gate

MCQ 3. A tertiary source is:

- a. Highly summarised (encyclopaedia, textbook index)
- b. A lab notebook
- c. A witness interview
- d. A packet capture

MCQ 4. Your own class survey is:

- a. Primary
- b. Tertiary
- c. Always secondary
- d. Not a source

MCQ 5. A newspaper report of that survey is:

- a. Secondary
- b. Primary always
- c. Tertiary always
- d. IoT

MCQ 6. Board papers want you to:

- a. Judge reliability, not just copy Google
- b. Use any blog
- c. Ignore dates
- d. Prefer unsigned forwards

Extra short questions

1. Define the three source types with one example each.
2. Which is a BIEK past paper for you as a student?
3. One reliability test.
4. Why tertiary is still useful.

Extra long question

1. For a project "screen time vs grades", list two primary, two secondary, one tertiary source and how you would check each. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Primary: experiment, interview, sensor. Secondary: review article, textbook chapter discussing others. Tertiary: encyclopaedia, library catalogue.
2. Primary evidence of the exam; or secondary if you only read a guide about it — say so.
3. Author, date, purpose, evidence, bias.
4. Fast overview and pointers to better sources — not the final citation for a claim.

Long

1. Primary: your survey, school lab logs. Secondary: a journalistic feature, a review paper. Tertiary: textbook definition of survey. Checks: sample size, date, conflict of interest.

Next lecture: 5.4 Impacts of Computing in Various Fields